# MTH2001

## UNIVERSITY OF EXETER

## COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

## MATHEMATICS

## January 2019

### Analysis

### Module Leader: Dr Jimmy Tseng

### Duration: 2 HOURS.

**Answer all questions.**

*Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.*

This is a **CLOSED NOTE** examination.

1. Consider the following subset of the real numbers

$$S = [-18, 1] \cup (3, \infty)$$

(a) Is the set open, closed, both open and closed, neither open nor closed? Justify your answer. (5)

(b) Determine $S^c = \mathbb{R} \setminus S$ and the set of interior points of $S^c$. (5)

[**10**]

2. Prove, using the $\epsilon$ and $\delta$ definition of a limit, that

$$\lim_{x \to 0} x \cos\left(\frac{1}{x^3}\right) = 0.$$

[**10**]

3. (a) Define what it means for a function $f : [a, b] \to \mathbb{R}$ to be Lipschitz continuous. (5)

(b) Obtain the smallest Lipschitz constant for the following function on its domain of definition

$$f : [0, 4] \to \mathbb{R}, \quad f(x) = (x + 1)e^{-x}.$$

(5)

[**10**]

4. (a) Let $n \in \mathbb{N}$, $f_n : [a, b] \to \mathbb{R}$ and $f : [a, b] \to \mathbb{R}$. Suppose $f_n(x) \to f(x)$ $\forall x \in [a, b]$. State suitable conditions for which it follows that

$$\lim_{n \to \infty} \int_a^b f_n(x) \mathrm{d}x = \int_a^b f(x) \mathrm{d}x.$$

(5)

(b) Hence or otherwise, evaluate the limit

$$\lim_{n \to \infty} \int_1^2 e^{-nx^2} \mathrm{d}x.$$

(5)

[**10**]

5. (a) Define what it means for a function $f : \mathbb{R}^n \to \mathbb{R}^m$ to be differentiable at $p \in \mathbb{R}^n$. (5)

   (b) Fix $b \in \mathbb{R}^n$ and let $f : \mathbb{R}^n \to \mathbb{R}$ be given by

$$f(x) = \langle x, b \rangle,$$

   where $\langle a, b \rangle$ denotes the inner product of $a$ and $b$.
   Show that $f$ is differentiable at each $p \in \mathbb{R}^n$ (5)

[**10**]

6. (a) Let $a, b$ be real numbers. State, using $\epsilon$ and $\delta$, what does it mean for a function $f : [a, b] \to \mathbb{R}$ to be continuous from the right at $a$. (5)

   (b) Let $f(x)$ be the function

$$f(x) = \frac{\cos\left(\frac{1}{x}\right) e^{x^5}}{x^4 \sin x}.$$

   Is it true that there exists $M > 0$ such that $|f(x)| \leq M$ for all $1 \leq x \leq 2$? Justify your answer. (10)

   (c) Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that there exist $a, b, x_0 \in \mathbb{R}$ with $a \leq x_0 \leq b$ such that $f(x) \geq f(x_0)$ for all $x \in (-\infty, a) \cup (b, \infty)$. Prove that there exists $m \in \mathbb{R}$ such that $m = \min\{f(x) | x \in \mathbb{R}\}$. (10)

[**25**]

7. Suppose $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \to \mathbb{R}$ be a bounded function. Let $P = (x_0, \ldots, x_k)$ be a partition of $[a, b]$ where $x_0 = a$ and $x_k = b$.

   (a) Define the lower and upper Riemann sums $L(f; P)$ and $U(f; P)$. (5)

   (b) Define the lower and upper Riemann integrals of $f$

$$\underline{\int_a^b} f(x)\,\mathrm{d}x, \quad \overline{\int_a^b} f(x)\,\mathrm{d}x,$$

   and state what it means for a function $f$ to be Riemann integrable. (10)

(c) Let $f : [0, 1] \to \mathbb{R}$ be the function defined by $f(x) = x^2$, and let $\hat{P} = (0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}, 1)$ be a partition. Use the previous parts of this question and partition $\hat{P}$, to prove that the function $f$ is Riemann integrable over the interval $[0, 1]$.
(You may use $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$ without proving it.) (10)

**[25]**